Matrices & Vectors

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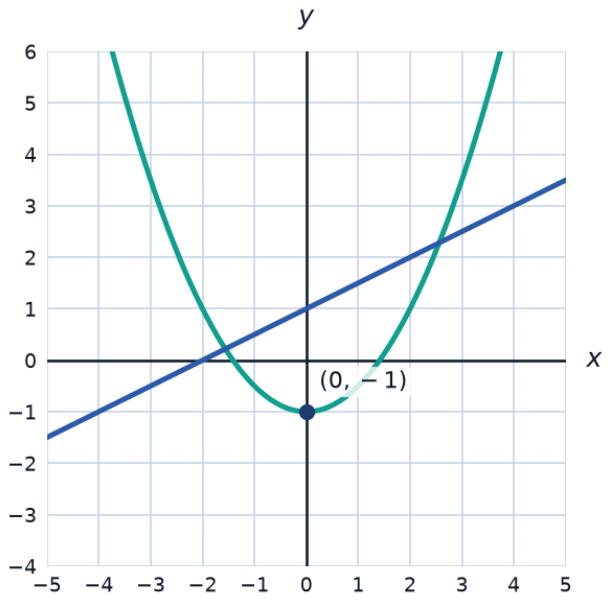
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Addition and scalar multiplication**

Let A be the matrix with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and B be the matrix with rows $\langle -1, 4\rangle$ and $\langle 0, 2\rangle$. Find $A + B$ and $3A$.

2 **[Foundational]** • **Target: Addition and scalar multiplication**

Using the same A and B, find $A - 2B$.



3 **[Foundational]** • **Target: Matrix multiplication**

Using A (rows $\langle 2, 1\rangle$, $\langle 3, -1\rangle$) and B (rows $\langle -1, 4\rangle$, $\langle 0, 2\rangle$) from above, find AB.

4 **[Foundational]** • **Target: Matrix multiplication**

Let A have rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and let $\mathbf{v} = \langle 2, -3\rangle$. Find $A\mathbf{v}$.

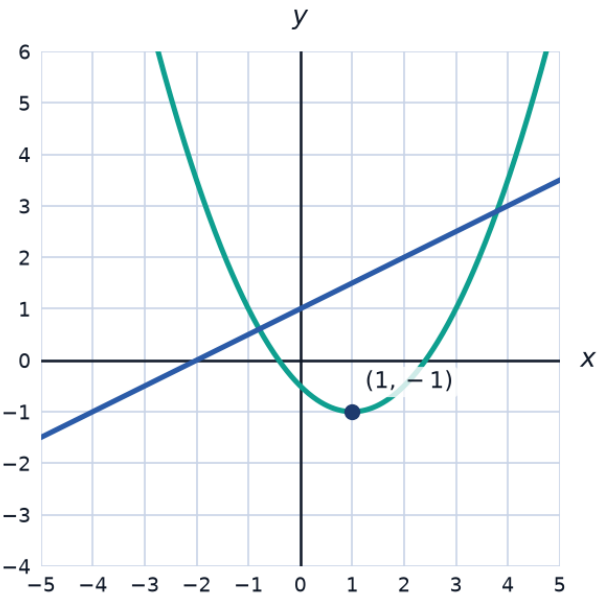
5 **[Foundational]** • **Target: Matrix multiplication**

Explain, without computing BA, why AB and BA need not be equal for 2×2 matrices.

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6 [Foundational] • Target: Determinant and inverse

Find $\det A$ for the matrix A with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$, and state whether A is invertible.



7 [Foundational] • Target: Determinant and inverse

Find A^{-1} for the matrix A with rows $\langle 2, 1\rangle$ and $\langle 3, -1\rangle$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Determinant and inverse

Without computing it, explain why the matrix with rows $\langle 4, 2\rangle$ and $\langle 6, 3\rangle$ has no inverse.

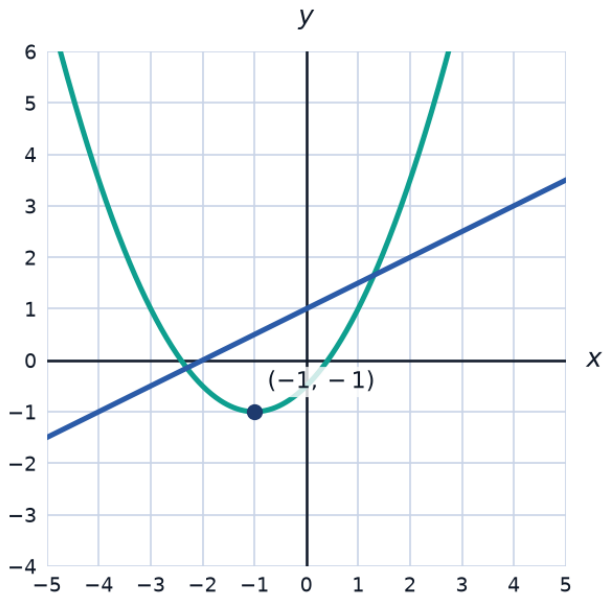
9 [Intermediate] • Target: Solving systems with the inverse matrix

Solve the system $3x - y = 8$ and $x + 2y = 5$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

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10 [Intermediate] • Target: Vector magnitude and dot product

For $\mathbf{u} = \langle 3, 4 \rangle$ and $\mathbf{w} = \langle -4, 3 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.



11 [Intermediate] • Target: Vector magnitude and dot product

What does $\mathbf{u} \cdot \mathbf{w} = 0$ tell you about $\mathbf{u}$ and $\mathbf{w}$ from the previous question?

12 [Intermediate] • Target: Angle between vectors

Find the angle between $\mathbf{p} = \langle 1, 0 \rangle$ and $\mathbf{q} = \langle 1, 1 \rangle$.

13 [Intermediate] • Target: Angle between vectors

Find the angle between $\langle 6, -8 \rangle$ and $\langle 3, 4 \rangle$, rounded to the nearest tenth of a degree.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the magnitude and direction angle θ (in degrees, $0^\circ \le \theta < 360^\circ$) of the vector $\mathbf{v} = \langle -5, 5\sqrt{3} \rangle$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

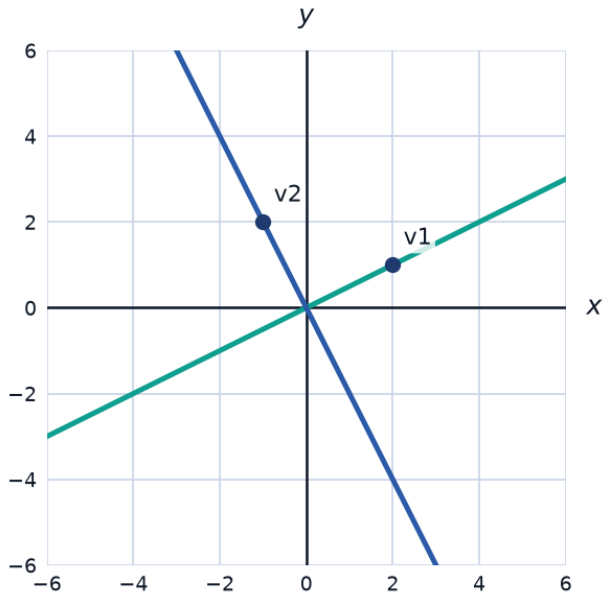
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

Find the angle θ between the vectors $\mathbf{a} = \langle 1, 2, 2 \rangle$ and $\mathbf{b} = \langle 2, -2, 1 \rangle$.

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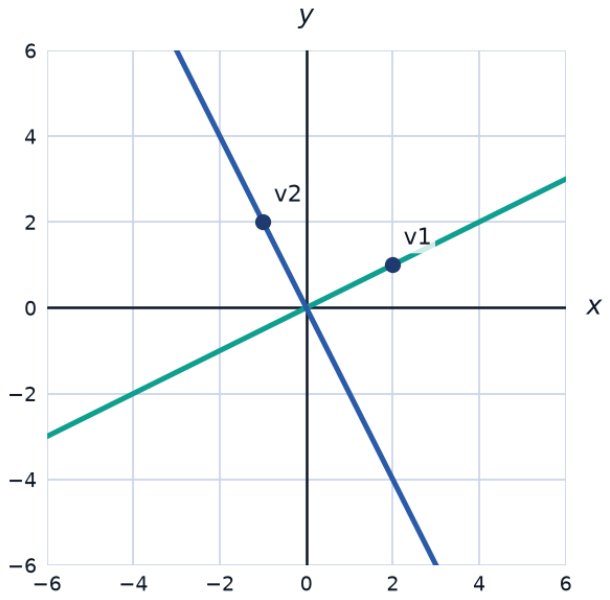
16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Sequences & Series

Find the inverse of the matrix $A = (4, 7; 1, 2)$.



17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Functions & Equations

Compute the matrix product AB where $A = (1, 3; -2, 4)$ and $B = (2, 0; 5, -1)$.



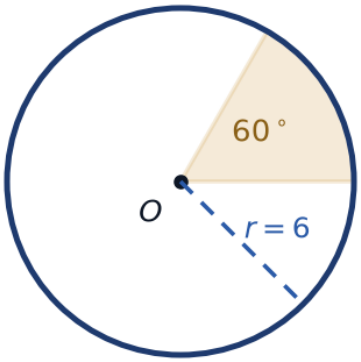
18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Calculate the determinant of the matrix $M = (7, 3; -2, 5)$.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Limits

Find the distance in the polar coordinate plane between the two points $P_1\left(4, \frac{\pi}{3}\right)$ and $P_2\left(7, \frac{2\pi}{3}\right)$.



20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Complex Numbers

Solve the system using matrix inversion: $4x + 7y = 11$ $x + 2y = 3$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Addition and scalar multiplication

Find $\det A$ for the matrix A with rows $\langle -2, 9\rangle$ and $\langle -5, -1\rangle$.

a. 19

b. 47

c. -43

d. 49

22 [Foundational] • Target: Matrix multiplication

Find $\det A$ for the matrix A with rows $\langle 2, -2\rangle$ and $\langle 9, -8\rangle$.

a. 5

b. -34

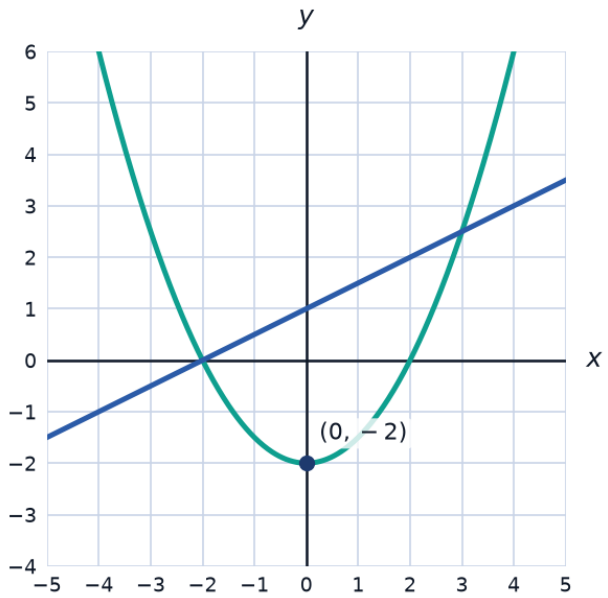
c. 2

d. 4

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23 [Foundational] • Target: Determinant and inverse

Find $\det A$ for the matrix A with rows $\langle -4, -8\rangle$ and $\langle -1, -3\rangle$.



a. 20

b. -20

c. 4

d. 6

24 [Foundational] • Target: Solving systems with the inverse matrix

Find $\det A$ for the matrix A with rows $\langle 0, -8\rangle$ and $\langle -4, 4\rangle$.

a. -30

b. 32

c. -29

d. -32

25 [Foundational] • Target: Vector magnitude and dot product

Find $\det A$ for the matrix A with rows $\langle -5, 0\rangle$ and $\langle -7, 8\rangle$.

a. -40

b. 35

c. -37

d. -38

26 [Foundational] • Target: Angle between vectors

Find $\det A$ for the matrix A with rows $\langle 7, -5\rangle$ and $\langle 6, -4\rangle$.

a. 2

b. 4

c. -58

d. 22

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find $\det A$ for the matrix A with rows $\langle -7, 9\rangle$ and $\langle 6, -7\rangle$.

a. -5

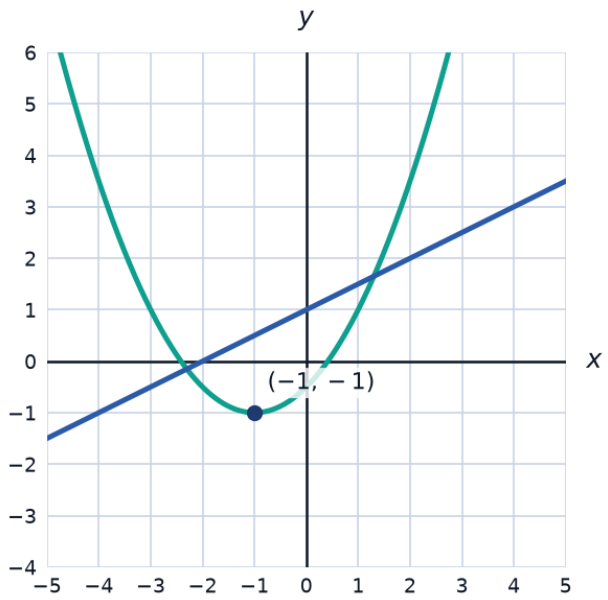
b. 103

c. -3

d. 21

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find $\det A$ for the matrix A with rows $\langle 5, 2\rangle$ and $\langle -9, -3\rangle$.



a. -33

b. 3

c. 5

d. -39

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Let A have rows $\langle -3, 7\rangle$ and $\langle 1, 3\rangle$, and let $\mathbf{v} = \langle 3, -5\rangle$. Find $A\mathbf{v}$.

a. $\langle -42, -12\rangle$

b. $\langle -12, -44\rangle$

c. $\langle -44, -12\rangle$

d. $\langle -44, -14\rangle$

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let A have rows $\langle 1, 1\rangle$ and $\langle 7, -7\rangle$, and let $\mathbf{v} = \langle 7, 7\rangle$. Find $A\mathbf{v}$.

a. $\langle 14, 0\rangle$

b. $\langle 0, 14\rangle$

c. $\langle 14, -2\rangle$

d. $\langle 16, 0\rangle$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Let A have rows $\langle -1, 6\rangle$ and $\langle 2, 2\rangle$, and let $\mathbf{v} = \langle -1, 5\rangle$. Find $A\mathbf{v}$.

a. $\langle 33, 8\rangle$

b. $\langle 31, 8\rangle$

c. $\langle 8, 31\rangle$

d. $\langle 31, 6\rangle$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let A have rows $\langle -7, -6\rangle$ and $\langle 8, 4\rangle$, and let $\mathbf{v} = \langle -5, -4\rangle$. Find $A\mathbf{v}$.

a. $\langle 61, -56\rangle$

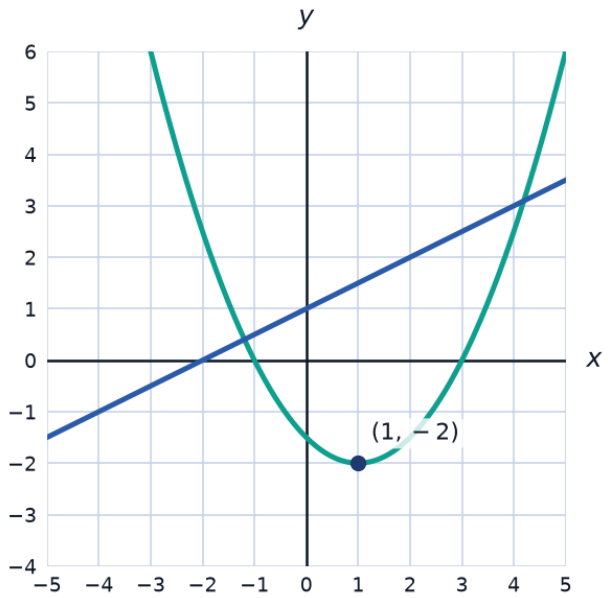
b. $\langle 59, -56\rangle$

c. $\langle 59, -58\rangle$

d. $\langle -56, 59\rangle$

33 [Intermediate] • Target: Addition and scalar multiplication

Let A have rows $\langle -2, -6\rangle$ and $\langle 7, -5\rangle$, and let $\mathbf{v} = \langle -9, -3\rangle$. Find $A\mathbf{v}$.



a. $\langle 38, -48\rangle$

b. $\langle 36, -48\rangle$

c. $\langle 36, -50\rangle$

d. $\langle -48, 36\rangle$

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34 [Intermediate] • Target: Matrix multiplication

Let A have rows $\langle -7, -2\rangle$ and $\langle 1, 0\rangle$, and let $\mathbf{v} = \langle 8, -1\rangle$. Find $A\mathbf{v}$.

a. $\langle -54, 6\rangle$

b. $\langle 8, -54\rangle$

c. $\langle -52, 8\rangle$

d. $\langle -54, 8\rangle$

35 [Intermediate] • Target: Determinant and inverse

Let A have rows $\langle 3, -2\rangle$ and $\langle -5, 1\rangle$, and let $\mathbf{v} = \langle 5, 5\rangle$. Find $A\mathbf{v}$.

a. $\langle 5, -22\rangle$

b. $\langle 7, -20\rangle$

c. $\langle 5, -20\rangle$

d. $\langle -20, 5\rangle$

36 [Intermediate] • Target: Solving systems with the inverse matrix

Find A^{-1} for the matrix A with rows $\langle 3, 1\rangle$ and $\langle -1, 0\rangle$.

a. $\langle 0, -1\rangle, \langle 1, 3\rangle$

b. $\langle 0, 1\rangle, \langle -1, 3\rangle$

c. $\langle 1, 3\rangle, \langle 0, -1\rangle$

d. $\langle 3, 1\rangle, \langle -1, 0\rangle$

37 [Intermediate] • Target: Vector magnitude and dot product

Find A^{-1} for the matrix A with rows $\langle 1, -2\rangle$ and $\langle 2, -5\rangle$.

a. $\langle 5, 2\rangle, \langle -2, -1\rangle$

b. $\langle 2, -1\rangle, \langle 5, -2\rangle$

c. $\langle 5, -2\rangle, \langle 2, -1\rangle$

d. $\langle 1, -2\rangle, \langle 2, -5\rangle$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Angle between vectors • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

Solve the system $6y = -30$ and $6x = 24$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle -5, 4\rangle$

b. $\langle 6, -5\rangle$

c. $\langle 4, -5\rangle$

d. $\langle 4, -7\rangle$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Limits

Solve the system $x + 3y = -15$ and $x + 2y = -8$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle 6, -7\rangle$

b. $\langle -7, 6\rangle$

c. $\langle 8, -7\rangle$

d. $\langle 6, -9\rangle$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Complex Numbers

Solve the system $x - y = 0$ and $-3x - 6y = -54$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle 9, 6\rangle$

b. $\langle 6, 4\rangle$

c. $\langle 6, 6\rangle$

d. $\langle 8, 6\rangle$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Sequences & Series

For $\mathbf{u} = \langle -20, -21\rangle$ and $\mathbf{v} = \langle 8, -15\rangle$, find $\cos\theta$ where θ is the angle between them, using $\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{155}{493}$

b. $-\frac{155}{493}$

c. $\frac{156}{493}$

d. $\frac{155}{46}$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Advanced Functions & Equations

For $\mathbf{u} = \langle 8, 15\rangle$ and $\mathbf{v} = \langle 7, 24\rangle$, find $\cos\theta$ where θ is the angle between them, using $\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{417}{425}$

b. $\frac{208}{21}$

c. $-\frac{416}{425}$

d. $\frac{416}{425}$

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Trigonometry & Polar Coordinates

For $\mathbf{u} = \langle -20, -21 \rangle$ and $\mathbf{v} = \langle 8, -15 \rangle$, find $\cos\theta$ where θ is the angle between them, using $\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{155}{493}$

b. $-\frac{155}{493}$

c. $\frac{156}{493}$

d. $\frac{155}{46}$

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Limits

If A is a 3×2 matrix and B is a 2×4 matrix, what are the dimensions of the product matrix AB?

a. 3×4

b. 2×2

c. 4×3

d. Multiplication is undefined

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Complex Numbers

For what value of x is the matrix $[[x, 3], [12, x]]$ singular?

a. ± 6

b. ± 4

c. ± 12

d. 0

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